LiDAR Point Cloud Denoising for Autonomous Driving Applications

Capstone Project Proposal

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Mentor Consent Form

I hereby agree to be the mentor of the following Capstone Project Team.

Project Title: LiDAR Point Cloud Denoising for Autonomous Driving Applications

Roll No.	**Name**	**Signature**

NAME of Mentor: .....

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NAME of Co-Mentor (if any): .....

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1 Project Overview

LiDAR (Light Detection and Ranging) sensors are widely used in autonomous driving, robotics, and smart surveillance systems to capture accurate 3D representations of the environment. However, raw LiDAR point cloud data is often corrupted by noise due to environmental factors such as rain, fog, dust, and sensor inaccuracies. This noise affects downstream tasks such as object detection, localization, and scene understanding.

The proposed capstone project focuses on designing a robust pipeline for denoising LiDAR point clouds. The system will be evaluated using standard benchmark datasets such as KITTI, nuScenes, and Waymo Open Dataset, which represent real-world driving and urban scenarios. The objective is to remove noise while preserving important geometric and structural information of objects.

The proposed solution aims to balance accuracy and computational efficiency, enabling its application in near real-time settings. Such a system is crucial for time-sensitive applications like autonomous vehicles and robotic navigation, where reliable perception of the environment is essential for safe and effective operation.

2 Need Analysis

The rapid growth of autonomous systems - including self-driving vehicles, UAVs, and service robots - has significantly increased reliance on LiDAR-based perception. These systems depend on accurate 3D point cloud data to perform critical tasks such as obstacle detection, path planning, and environment mapping. LiDAR has become the sensor of choice for Level 3 and Level 4 autonomy, as no other single sensor provides the dense, range-accurate 3D point cloud that collision avoidance and path planning require.

Real-world LiDAR data is, however, consistently corrupted by noise arising from three distinct sources. *Environmental conditions* such as rain, fog, and dust cause laser pulses to backscatter off suspended particles before reaching their intended targets, generating spurious phantom points at incorrect distances. *Hardware limitations* - including multi-echo interference between co-mounted sensors, intensity calibration errors, and vibration-induced jitter - introduce systematic outliers that geometric filters struggle to distinguish from valid returns. *Motion artefacts* arise because a spinning LiDAR takes approximately 100 ms to complete one scan; at highway speeds, this displacement distorts the shape of nearby objects along the direction of travel.

The downstream consequences are serious. Object detectors trained on clean data degrade significantly on noisy inputs, ICP-based localisation diverges in the presence of outliers, and SLAM pipelines can fail entirely under heavy environmental noise. The sensitivity of deep 3D perception models to point-level data quality is further underscored by the adversarial attack literature: research has shown that even imperceptible, deliberately crafted perturbations added to LiDAR point clouds are sufficient to cause state-of-the-art detectors to fail catastrophically across multiple architectures [11, 12]. If deliberate micro-perturbations can break detectors so reliably, naturally occurring noise - which is orders of magnitude less controlled - poses an equally serious threat to perception reliability. Existing remedies fall into two families, neither of which is fully satisfactory. Statistical filters (SOR, ROR) are fast but apply a global threshold that erodes fine geometric detail and cannot distinguish structure from noise semantically [5]. Deep learning methods are more accurate but require GPU resources, large labelled datasets, and often fail to generalise across weather types or sensor configurations [6]. A solution that is simultaneously accurate, computationally efficient, and generalisable remains an open problem - and is precisely the gap this project aims to address.

3 Literature Survey

LiDAR point cloud denoising has been an active research area for over a decade, driven by the growing deployment of autonomous systems in uncontrolled outdoor environments. The literature can be broadly organised into four generations of approaches: classical statistical filters, projection-based convolutional methods, native 3D deep learning architectures, and generative models. Each generation has addressed limitations of the previous one, while introducing its own trade-offs between accuracy, computational cost, and generalisability. This survey examines each family in turn and contextualises their relevance to the proposed project.

3.1 Classical Statistical and Geometric Methods

The earliest and most widely deployed denoising techniques operate directly on point neighbourhood statistics without any learned components. Statistical Outlier Removal (SOR), introduced as part of the Point Cloud Library (PCL) [7], computes for each point the mean distance $\bar{d}$ to its k nearest neighbours and removes any point whose $\bar{d}$ exceeds $\mu + \alpha\sigma$, where μ and σ are the global mean and standard deviation of all neighbourhood distances and α is a user-set multiplier. Radius Outlier Removal (ROR) takes a complementary approach: a point is removed if it has fewer than a minimum count of neighbours within a fixed radius r. Both methods run in $\mathcal{O}(n \log n)$ time using a k-d tree, making them practical even on embedded hardware.

Despite their efficiency, classical filters carry well-known failure modes. Because SOR applies a single global threshold, it is unable to distinguish genuine sparse returns - such as the few points returned by a distant pedestrian or a low-reflectivity road sign - from sparse noise clusters. Tuning k and α for dense near-field noise simultaneously causes over-smoothing of thin or distant structures [6]. Bilateral filtering and Moving Least Squares (MLS) surface reconstruction offer improved geometric fidelity by weighting neighbour contributions by spatial proximity or fitting local polynomials to the surface, but both introduce significantly higher computational cost and assume locally smooth geometry that does not hold at object boundaries or sharp edges.

The collective limitation of all classical approaches is their lack of semantic awareness: they operate on geometry alone and have no mechanism to distinguish a noisy but structurally meaningful region from a clean but uninformative one.

3.2 Projection-Based Convolutional Methods

A natural bridge between classical signal processing and modern deep learning is to project the irregular point cloud onto a regular 2D representation and apply standard convolutional neural networks (CNNs). Range image projection maps each 3D point to a pixel in a cylindrical panorama by encoding its depth, height, and reflectance. This preserves neighbourhood structure and allows reuse of mature 2D CNN architectures without modification.

Heinzler et al. [6] proposed one of the first CNN-based weather denoising pipelines operating on range images. Their model was trained on a labelled dataset of LiDAR scans collected under controlled rain and fog conditions, with each point annotated as *clear* or *precipitation*. Using a U-Net-style encoder-decoder, the network achieved substantially higher precision-recall for noise classification than SOR or ROR on their proprietary benchmark. A key finding was that convolutional features learned to exploit the spatial coherence of precipitation clusters - something purely statistical filters cannot do. However, the range image projection introduces an inherent artefact: points that lie at the same azimuth and elevation but at different ranges are mapped to the same pixel, causing occlusion losses. The cylindrical projection is also specific to spinning LiDARs and does not generalise to solid-state or MEMS-based sensors with irregular scan patterns.

3.3 Native 3D Deep Learning Architectures

The limitations of projection motivated architectures that operate directly in 3D, treating the point cloud as an unordered set. PointNet [1] was the foundational work in this space: it applied independent multi-layer perceptrons (MLPs) to each point, followed by a symmetric aggregation function (max-pooling) to produce a global feature vector invariant to point permutation. While PointNet proved highly effective for classification and part segmentation, its per-point processing is local and does not capture fine neighbourhood geometry. PointNet++ [2] addressed this by introducing a hierarchical grouping mechanism - farthest point sampling followed by ball-query neighbourhood aggregation - that learns features at multiple spatial scales, substantially improving performance on tasks requiring local geometric understanding such as surface normal estimation and denoising.

Graph-based methods represent an alternative approach to capturing local structure. DGCNN [8] constructs a k-nearest-neighbour graph in feature space at each layer and

applies edge convolutions that aggregate relative feature differences from neighbours, enabling the network to adapt its receptive field dynamically as features evolve. Applied to denoising, graph convolutions can distinguish noise clusters - which typically form isolated subgraphs - from surface points that participate in large, well-connected components.

Transformer architectures have more recently been applied to point clouds. Point Transformer [9] replaces graph convolution with a localised self-attention operator: for each point, attention weights are computed over its k nearest neighbours using relative positional encodings, and the output is a weighted sum of neighbour features. This formulation is permutation-invariant by construction, captures richer geometric context than edge convolutions, and achieves state-of-the-art results on ModelNet40 classification and ShapeNet part segmentation. For outdoor denoising, the challenge is computational: a full self-attention over $N \sim 100{,}000$ points requires $\mathcal{O}(N^2)$ memory, necessitating local windowing or strided sampling strategies that introduce their own boundary artefacts.

Score-based denoising methods, inspired by the connection between Langevin dynamics and score matching, train a network to estimate the gradient of the log-density of the clean point distribution and iteratively move noisy points toward regions of higher density. This approach achieves competitive results on synthetic ShapeNet noise benchmarks but has not been demonstrated on large-scale outdoor driving data.

3.4 Generative and Diffusion-Based Models

The most recent direction in point cloud denoising draws from generative modelling. Denoising Diffusion Probabilistic Models (DDPMs) [10], originally developed for image synthesis, define a forward process that gradually adds Gaussian noise to data and a reverse process that learns to undo it. This framework has been adapted to point clouds by operating on point coordinates directly, demonstrating high-quality generation of clean shapes from heavily corrupted inputs. The key advantage over discriminative models is that the diffusion prior implicitly regularises geometry - the reverse process can reconstruct plausible surface structure even in regions with very few clean points.

However, diffusion models carry a fundamental inference-time cost: each denoising step requires a full forward pass through a large neural network, and competitive quality typically requires hundreds to thousands of steps. Accelerated sampling strategies reduce this gap but per-frame latency of several seconds on a GPU still renders these models

incompatible with 10 Hz real-time requirements. Real-time deployment of diffusion-based denoising therefore remains an open research problem.

3.5 Comparison and Research Gap

Table 1 summarises the key methods discussed above across three axes relevant to the proposed project.

Table 1: Comparison of LiDAR point cloud denoising methods.

Method	**Accuracy**	**Real-Time**	**Generalisation**
SOR / ROR [7]	Low	Yes	High
CNN (range image) [6]	High	Yes (GPU)	Weather-specific
PointNet++ [2]	High	Marginal	Moderate
DGCNN [8]	High	No	Moderate
Point Transformer [9]	Very High	No	Moderate
Diffusion Models [10]	Very High	No	High

As the table makes clear, no single method simultaneously achieves high accuracy, real-time throughput, and broad generalisability. Classical methods are fast but inaccurate; deep learning methods are accurate but slow or hardware-dependent; generative models offer the best geometric fidelity but are orders of magnitude too slow for embedded deployment.

Three specific gaps emerge from this survey. First, there is no lightweight architecture that has been rigorously evaluated on large-scale outdoor driving data under multiple noise types simultaneously - most deep learning work is benchmarked on synthetic ShapeNet objects, which have very different geometric properties to sparse, rotating-LiDAR scans of road scenes. Second, existing methods treat denoising in isolation: they optimise reconstruction metrics such as Chamfer Distance without measuring whether improved point cloud quality actually translates into better downstream object detection accuracy, which is ultimately the metric that matters for autonomous driving. Third, real-time feasibility is rarely reported under consistent hardware constraints, making it difficult to compare methods fairly for deployment.

The proposed project targets precisely these gaps. By training and evaluating on KITTI and nuScenes, benchmarking against both reconstruction metrics and detection accuracy, and explicitly profiling throughput against the 10 Hz automotive LiDAR constraint, this

work addresses limitations that individually motivated prior methods but have not yet been treated as a unified design objective.

4 Objectives

The main objectives of the proposed project are:

1. To study the characteristics of noise in LiDAR point cloud datasets.
2. To design a framework for denoising LiDAR point cloud data while preserving geometric and structural information.
3. To evaluate the performance of the proposed approach using standard metrics such as reconstruction error (Chamfer Distance, Hausdorff Distance) and improvement in downstream object detection accuracy.
4. To optimise the proposed approach for near real-time processing, making it suitable for deployment in autonomous systems.

5 Methodology

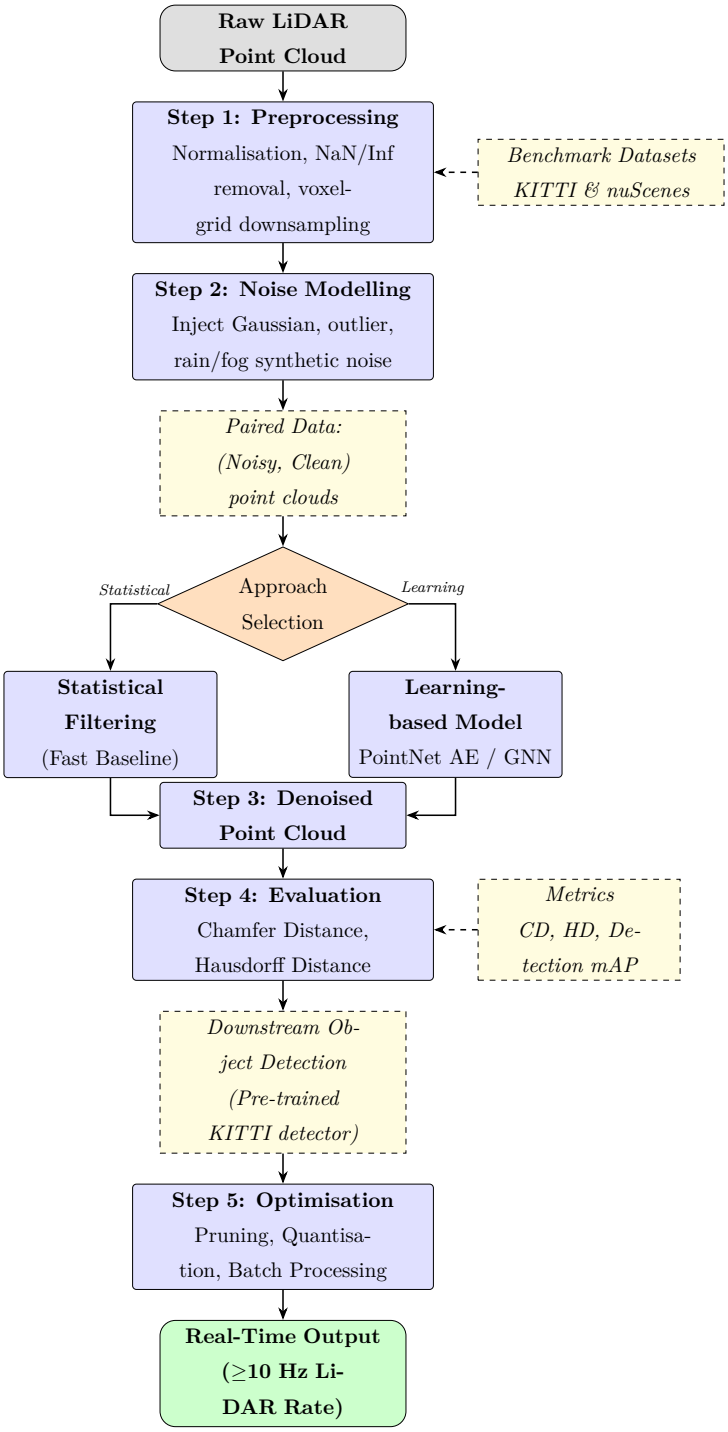


Figure 1: Logical flow diagram of the proposed LiDAR point cloud denoising system.

The proposed methodology involves a stepwise approach for designing and evaluating a LiDAR point cloud denoising system.

Step 1: Dataset Study and Preprocessing

Benchmark datasets - KITTI [3] and nuScenes [4] - will be studied to understand the characteristics of real-world LiDAR data. Input point clouds will undergo preprocessing steps such as normalisation, removal of invalid (NaN/Inf) points, and voxel-grid downsampling to reduce computational overhead.

Step 2: Noise Modelling

Synthetic noise (Gaussian, random outliers, rain/fog simulation) will be injected into clean point clouds to create paired (noisy, clean) training data. This supervised training signal enables quantitative evaluation of denoising quality.

Step 3: Denoising Pipeline Design

A denoising technique will be selected and implemented. The approach may include: (a) statistical filtering as a fast baseline, or (b) a learning-based model (e.g., PointNet-based autoencoder or lightweight graph neural network) depending on feasibility. The goal is to remove noise while preserving the geometric structure of objects in the scene.

Step 4: Evaluation

Performance will be assessed by comparing original and processed point clouds using metrics such as Chamfer Distance and Hausdorff Distance. The impact on downstream object detection accuracy (using a pre-trained detector on KITTI) will also be measured.

Step 5: Optimisation for Real-Time Performance

The pipeline will be profiled and optimised (e.g., model pruning, quantisation, or batch processing) to determine suitability for near real-time operation, targeting throughput compatible with typical LiDAR scan rates (10 Hz).

6 Project Requirements

Technical Requirements

- **Programming Language:** Python
- **Frameworks:** PyTorch / TensorFlow for model development and training
- **Libraries:** Open3D and PCL for point cloud processing; NumPy for numerical operations
- **Datasets:** KITTI and nuScenes (publicly available LiDAR benchmarks)
- **Compute:** GPU access (minimum 8 GB VRAM recommended) for training deep learning models

Hardware Requirements

- **LiDAR Sensor:** YDLIDAR X2 2D LiDAR sensor for live data collection and pipeline testing
- **Interface:** USB / Ethernet connection to host machine
- **Mount:** Tripod or fixed stand for stable scanning
- **Compute Platform:** Laptop or desktop (primary development); Raspberry Pi for lightweight embedded inference testing

7 Work Plan

The project will be executed over a period of six months as outlined in Table 2 and the Gantt chart in Figure 2.

Table 2: Six-month work plan for the capstone project.

Month	Activity	Milestone
1	Literature survey and problem understanding	Survey report
2	Dataset study (KITTI, nuScenes) and noise modelling	Dataset ready
3	Selection and initial implementation of denoising technique	Baseline result
4	Design and refinement of complete pipeline	Working pipeline
5	Evaluation, analysis, and optimisation	Evaluation report
6	Documentation and final report preparation	Final report

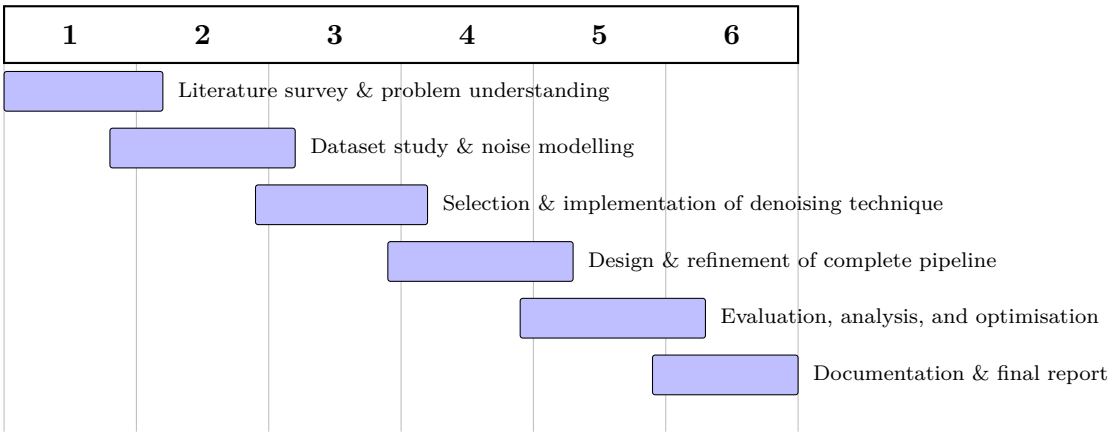


Figure 2: Gantt chart illustrating the six-month project timeline and milestones.

8 Project Outcomes & Individual Roles

Expected Outcomes

- A well-defined LiDAR point cloud denoising framework (code and documentation).
- Performance comparison of different denoising approaches (statistical vs. learning-based).
- Analysis of the impact of denoising on downstream object detection accuracy.
- Final project report and presentation.

Individual Roles

Table 3: Individual roles and responsibilities.

Name	**Roll No.**	**Responsibilities**
	102315219	Literature review
	102303537	Dataset analysis
	102303640	Methodology design and implementation
	102303703	Evaluation, benchmarking, and optimisation
	102315247	Documentation and report preparation

9 Course Subjects

The successful execution of this project will require knowledge from the following subjects:

- **Machine Learning** - for understanding and implementing learning-based denoising models.
- **Computer Vision** - for 3D scene understanding, object detection, and point cloud processing.
- **Data Structures and Algorithms** - for efficient neighbourhood search (k-d tree, octree) and pipeline design.
- **Linear Algebra** - for geometric transformations and matrix operations on point cloud data.
- **Probability and Statistics** - for statistical filtering techniques and performance metric analysis.
- **Parallel Computing** (*optional*) - for GPU-accelerated processing to achieve near real-time performance.

10 References

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